

AN AUTOMATED STOCHASTIC VOLATILITY MODEL PARAMETER CALIBRATION ROUTINE WITH OPTIONAL JUMPS

RADU BRICIU

ABSTRACT. We specify a routine that calibrates the five canonical Heston [6] parameters $\Phi = (\theta, \kappa, \eta, \rho, v_0)$ to a day of CBOE S&P 500 (SPX) intraday option trades. The routine performs *one* calibration per trading day over a pooled, moneyness-normalised implied-volatility surface, and is framed throughout as a box-constrained nonlinear least-squares problem solved from multiple seeds. We make explicit every modelling choice: the volume-weighted reference spot S_{ref} , the moneyness-normalised and grid-snapped strike K^* , the surface-coverage requirements, the configurable in-engine objective (defaulting to implied-volatility “IV-space” residuals), the IV-space restart selection and acceptance gate, and the boundary-pegging rejection rule. The same routine optionally fits the Bates [2] extension, which adds Merton lognormal jumps to the Heston dynamics, selected by a run-time flag and calibrated over the identical surface and gate. We keep Heston as the default throughout and present Bates as the eight-parameter generalisation it specialises from.

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1. The model

1.1. Basis. Under the risk-neutral measure, with constant risk-free rate r and continuous dividend yield g , the log-price $X_t = \ln S_t$ and instantaneous variance v_t of the SPX follow the Heston [6] dynamics

$$dX_t = (r - g - \frac{1}{2}v_t) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t), \quad (1)$$

$$dv_t = \kappa(\theta - v_t) dt + \eta\sqrt{v_t} dW_t, \quad (2)$$

where W, B are independent Brownian motions, so $\text{corr}(dX_t, dv_t) = \rho dt$. The parameter vector is

$$\Phi = (\theta, \kappa, \eta, \rho, v_0) \in \mathbb{R}^5, \quad (3)$$

with $\theta > 0$ the long-run variance, $\kappa > 0$ the mean-reversion speed, $\eta > 0$ the volatility of variance (QuantLib's `sigma`), $\rho \in [-1, 1]$ the leverage correlation, and $v_0 > 0$ the initial variance. The Feller quantity [5]

$$\mathcal{F}(\Phi) = 2\kappa\theta - \eta^2 \quad (4)$$

is reported per fit; $\mathcal{F}(\Phi) \geq 0$ guarantees $v_t > 0$ almost surely. It is not imposed as a constraint (short-tenor SPX fits routinely violate it without being otherwise unreasonable; see §6).

1.2. Jumps. The Bates [2] model extends Heston with Merton [8] lognormal jumps in the log-price. The variance dynamics (2) are unchanged. The level equation (1) gains a compound-Poisson jump term:

$$dX_t = (r - g - \frac{1}{2}v_t - \lambda\xi) dt + \sqrt{v_t} (\rho dW_t + \sqrt{1 - \rho^2} dB_t) + dJ_t. \quad (5)$$

Here $J_t = \sum_{n=1}^{N_t} Y_n$ is a compound-Poisson process: N_t is a Poisson process of intensity $\lambda \geq 0$ (expected jumps per year), independent of W and B , and the log-jump sizes $Y_n \sim \mathcal{N}(\nu, \delta^2)$ are i.i.d. The compensator

$$\xi = \mathbb{E}[e^Y - 1] = e^{\nu + \delta^2/2} - 1 \quad (6)$$

offsets the jump drift so the discounted price stays a martingale, leaving the level drift $r - g$ in expectation. The augmented parameter vector is

$$\Theta = (\theta, \kappa, \eta, \rho, v_0, \lambda, \nu, \delta) \in \mathbb{R}^8, \quad (7)$$

the five Heston parameters of (1)–(2) plus jump intensity $\lambda \geq 0$, mean log-jump ν , and log-jump volatility $\delta > 0$. At $\lambda = 0$ the jump term vanishes and (5) collapses to the Heston level (1), so Heston is the $\lambda = 0$ face of the Bates parameter space. The model is selected at run time by a flag; the Heston fit is the default. The Feller quantity (4) is built from the diffusive parameters alone, so jumps do not enter it; it is reported under both models.

1.3. Pricing operators. Fix a valuation with maturity $\tau > 0$ (in years), strike K , and type $w \in \{+1, -1\}$ (+1 call, -1 put). Write the forward and discount factor under flat curves as

$$F(S, \tau) = S e^{(r-g)\tau}, \quad D(\tau) = e^{-r\tau}. \quad (8)$$

Black price. Black [3] and Black and Scholes [4], with total standard deviation $\Sigma = \sigma\sqrt{\tau}$,

$$C_B(\sigma; S, K, \tau, w) = w D(\tau) \left[F \mathcal{N}(wd_+) - K \mathcal{N}(wd_-) \right], \quad (9)$$

$$d_{\pm} = \frac{\ln(F/K)}{\Sigma} \pm \frac{1}{2}\Sigma,$$

where $\mathcal{N}$ is the standard normal CDF and $F = F(S, \tau)$.

Heston price. The Heston model price $C_H(\Phi; S, K, \tau, w)$ is the semi-analytic value

$$C_H(\Phi; S, K, \tau, +1) = S e^{-g\tau} P_1 - K e^{-r\tau} P_2, \quad (10)$$

$$C_H(\cdot, -1) = C_H(\cdot, +1) - S e^{-g\tau} + K e^{-r\tau},$$

with the in-the-money probabilities, for $j \in \{1, 2\}$,

$$P_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left[\frac{e^{-iu \ln K} f_j(u; \Phi, \tau, \ln F)}{iu} \right] du, \quad (11)$$

and characteristic functions $f_j = \exp(C_j + D_j v_0 + iu \ln F)$ given by the standard coefficients

$$C_j = \frac{\kappa\theta}{\eta^2} \left[(b_j - \rho\eta iu + d_j)\tau - 2 \ln \frac{1 - \gamma_j e^{d_j\tau}}{1 - \gamma_j} \right], \quad (12)$$

$$D_j = \frac{b_j - \rho\eta iu + d_j}{\eta^2} \frac{1 - e^{d_j\tau}}{1 - \gamma_j e^{d_j\tau}},$$

$$d_j = \sqrt{(\rho\eta iu - b_j)^2 - \eta^2(2\mu_j iu - u^2)}, \quad \gamma_j = \frac{b_j - \rho\eta iu + d_j}{b_j - \rho\eta iu - d_j}, \quad (13)$$

where $\mu_1 = \frac{1}{2}$, $\mu_2 = -\frac{1}{2}$, $b_1 = \kappa - \rho\eta$, $b_2 = \kappa$. In code this integral is evaluated by the QuantLib [1] `AnalyticHestonEngine` (which uses a branch-stable formulation of (11)); equations (10)–(11) define the operator it realises.

Bates price. The Bates price $C_{\text{Bates}}(\Theta; S, K, \tau, w)$ has the same form as the Heston price (10)–(11), with each characteristic function f_j multiplied by the Merton lognormal jump factor of §1.2:

$$f_j^B(u) = f_j(u) e^{\tau \psi_j(u)}, \quad \psi_j(u) = \lambda \left(e^{iu_j \nu - \frac{1}{2} \delta^2 u_j^2} - 1 \right) - iu_j \lambda \xi, \quad (14)$$

with $u_2 = u$, $u_1 = u - i$ (the same measure shift that distinguishes f_1 from f_2), and $\xi = e^{\nu + \delta^2/2} - 1$ the compensator (6). The exponent ψ_j is the Merton log-jump cumulant, compensated by $-iu_j \lambda \xi$ so the drift correction of (5) carries through. At $\lambda = 0$ the factor is 1 and C_{Bates} reduces to C_H . In code the integral is evaluated by the QuantLib `BatesEngine`, the jump-augmented counterpart of the `AnalyticHestonEngine`.

Implied volatility. The Black-implied volatility of a price C is the inverse of (9) in its first argument,

$$\mathcal{I}(C; S, K, \tau, w) = \{ \sigma > 0 : C_B(\sigma; S, K, \tau, w) = C \}, \quad (15)$$

computed numerically (QuantLib `blackFormulaImpliedStdDev` on the forward (8), returning Σ , then $\sigma = \Sigma/\sqrt{\tau}$). This operator defines the IV-space residuals of §5.

2. Data and per-day notation

The pipeline ingests CBOE trade-level files, one per trading day d , restricted upstream to out-of-the-money (OTM) quotes. A day is a set of trade records indexed by i , each carrying

$$(t_i, K_i, w_i, q_i, P_i, \sigma_i, S_i, \tau_i), \quad (16)$$

being respectively the quote timestamp, strike, type $w_i \in \{+1, -1\}$, traded size, trade price, trade-implied volatility, the underlying level *at trade time* t_i , and the days to maturity. Maturities enter the engine as τ_i in years under Actual/365, i.e. $T_i = \tau_i/365$.

The day is rate-stamped with $r = r(d)$ and $g = g(d)$, the most recent risk-free and dividend rates on or before d (a flat-forward curve for each; under flat curves the evaluation date is immaterial to T_i , so it is set to the QuantLib “today”).

Pre-filter. We keep only economically usable, Heston-appropriate quotes. Write the ratio moneyness $\varrho_i = e^{-|\ln(K_i/S_i)|} \in (0, 1]$ (it equals 1 at the money and falls as the strike moves out). The kept set is

$$\mathcal{D} = \{i : \sigma_i > 0, \tau_{\min} \leq \tau_i \leq \tau_{\max}, \underline{m} < \varrho_i < \bar{m}\}. \quad (17)$$

The maturity floor $\tau_{\min}$ (Table 1) drops ultra-short maturities, which the diffusive Heston smile fits poorly and which otherwise drive η, κ to Feller-violating extremes that pollute the pooled fit; the ceiling $\tau_{\max}$ drops thin, stale long tenors. The cutoff $\bar{m}$ removes the near-ATM rows, and the floor $\underline{m}$ removes the deep-OTM lottery-ticket tail ($\varrho_i < \underline{m}$, roughly $|\ln(K_i/S_i)| > 0.51$), whose extreme prices peg the fit to its bounds. Flooring that tail recovers acceptance and tightens the fit while keeping the full tradeable wing.

3. Handling intraday spot movement

A Heston calibration is anchored to a *single* spot S , yet the trades in $\mathcal{D}$ occur across an intraday range of underlying levels $\{S_i\}$. We reconcile the two with one reference spot and a moneyness re-striking.

Reference spot. Define the volume-weighted reference spot

$$S_{\text{ref}} = \frac{\sum_{i \in \mathcal{D}} q_i S_i}{\sum_{i \in \mathcal{D}} q_i}. \quad (18)$$

The intraday range and its flag are

$$R = \frac{\max_i S_i}{\min_i S_i} - 1, \quad \text{HIGH_MOVE} = \mathbf{1}\{R > R_{\max}\}. \quad (19)$$

A flagged day ($R > R_{\max}$) is still calibrated and written, but the normalisation below is recorded as strained (its S_{ref} should be treated with suspicion).

Moneyness normalisation and grid snap. Each trade retains its moneyness at trade time and is re-struck to the reference spot, then snapped to the SPX strike grid of width Δ :

$$m_i = \frac{K_i}{S_i}, \quad K_i^* = \Delta \cdot \text{round}\left(\frac{m_i S_{\text{ref}}}{\Delta}\right). \quad (20)$$

The rationale is sticky-moneyness: over a session the implied-volatility smile is approximately stationary in K/S rather than in the absolute strike. Re-striking by (20) therefore re-centres the whole day onto S_{ref} so that trades executed at

different S_i but the *same* moneyness share a common surface column. The grid snap collapses the continuum of K_i^* values (one per trade) back onto a clean strike lattice. Pricing diagnostics (§7) nonetheless use each contract’s *original* (S_i, K_i) , since the calibrated parameters are spot-independent.

4. Surface construction

From the normalised trades we build a partial $K^* \times \tau$ implied-volatility surface, selecting the most liquid quotes nearest the money. The trades entering this stage are already the pre-filtered set $\mathcal{D}$, so every candidate strike sits in the OTM moneyness band $\underline{m} < \varrho_i < \bar{m}$ of (17). That band fixes the reach of each wing: the cutoff $\bar{m}$ sets the inner edge, the strikes closest to the money, and the floor $\underline{m}$ sets the outer edge, where the deep-OTM tail is dropped. “Nearest the money” below therefore means just inside $\bar{m}$, not at the money itself.

Maturity selection. A maturity enters the surface only if it carries both wings. For each distinct maturity τ present in $\mathcal{D}$, let $n_c(\tau)$ and $n_p(\tau)$ be the number of distinct K^* on its call and put wings, and let $V(\tau) = \sum_{i \in \mathcal{D}: \tau_i = \tau} q_i$ be its traded volume. Call τ *admissible* when each wing carries at least $\underline{n}_k$ distinct strikes. Keep the $\bar{N}_t$ highest-volume admissible maturities,

$$\mathcal{T} = \text{top-}\bar{N}_t \text{ by } V(\tau) \text{ among } \{\tau : n_c(\tau) \geq \underline{n}_k, n_p(\tau) \geq \underline{n}_k\}. \quad (21)$$

The wing test precedes the volume cap. A maturity with a one-sided or single-strike wing is discarded before the cap, rather than consuming one of its $\bar{N}_t$ slots. The kept set falls below $\bar{N}_t$ only when fewer than $\bar{N}_t$ maturities are admissible.

Strike selection (per maturity, per wing). Within each kept maturity $\tau \in \mathcal{T}$, separate calls ($w_i = +1$, OTM with $K_i^* \gtrsim S_{\text{ref}}$) from puts ($w_i = -1$, OTM with $K_i^* \lesssim S_{\text{ref}}$) and keep the $\bar{n}_k$ nearest-the-money distinct strikes on each wing:

$$\mathcal{K}_\tau = \underbrace{\{\bar{n}_k \text{ largest } K^* \text{ among puts at } \tau\}}_{\text{put wing}} \cup \underbrace{\{\bar{n}_k \text{ smallest } K^* \text{ among calls at } \tau\}}_{\text{call wing}}. \quad (22)$$

The $\bar{n}_k$ cap binds only where a wing carries more than $\bar{n}_k$ distinct strikes inside the band; otherwise the wing’s outer edge is set by the floor $\underline{m}$ alone.

Pivot to a surface. For each retained cell (k, τ) with $k \in \mathcal{K}_\tau$, set the surface entry to the implied volatility of the *highest-volume* trade in that cell, the one with the largest traded size,

$$\hat{\sigma}_{k,\tau} = \sigma_{i^*}, \quad i^* = \arg \max_{i: K_i^* = k, \tau_i = \tau} q_i, \quad (23)$$

and leave it missing otherwise. When several trades tie at the maximum size the pick is arbitrary. Let

$$\mathcal{N} = \{(k, \tau) : \hat{\sigma}_{k,\tau} \text{ defined}\} \quad (24)$$

be the set of populated cells; these become the calibration targets (helpers).

Coverage requirement. To keep the five-parameter fit identified we require a genuinely two-dimensional surface:

$$|\{\tau : (\cdot, \tau) \in \mathcal{N}\}| \geq M_t, \quad |\{k : (k, \cdot) \in \mathcal{N}\}| \geq M_k, \quad |\mathcal{N}| \geq M_{\text{cell}}. \quad (25)$$

A day failing (25) (or with no usable maturities) is dropped: it contributes no calibration and clears its stale diagnostics.

5. The optimization problem

The routine calibrates one parameter vector per day. Under the default Heston model this is Φ of (3) with $\dim \Phi = 5$; under Bates it is Θ of (7) with $\dim \Theta = 8$. The two are nested. They share their first $\dim \Phi$ coordinates, the diffusive parameters $(\theta, \kappa, \eta, \rho, v_0)$, and Θ appends the jump triple (λ, ν, δ) . Setting $\lambda = 0$ collapses the jump term (5) and recovers Φ , so Heston is the $\lambda = 0$ face of Bates. We write Ψ for the active vector, either Φ or Θ , and $C^{\text{mod}}(\Psi; \cdot)$ for the active model price: the Heston price C_H (10) when $\Psi = \Phi$, and the Bates price C_{Bates} (14) when $\Psi = \Theta$. Everything below is written in Ψ and holds for either model. The only place the model enters is the box dimension $\dim \Psi$ (26) and, through it, the bounds tables.

Let $\mathcal{N}$ index the targets with market data $\{(\hat{\sigma}_n, K_n^*, T_n, w_n)\}_{n \in \mathcal{N}}$, all priced at the single spot S_{ref} . Define the feasible box

$$\mathcal{B} = \prod_{j=1}^{\dim \Psi} [\ell_j, h_j] \subset \mathbb{R}^{\dim \Psi}, \quad (26)$$

with (ℓ, h) in the `model.params()` order of Table 2: the five Heston coordinates, extended by the three jump bounds when $\Psi = \Theta$. The λ floor is exactly 0, so the Bates box contains the Heston model on its $\lambda = 0$ face.

5.1. Residuals. Two residual systems are defined: an IV-space system and a price-space system. The in-engine objective is configurable. It defaults to the IV-space system (28) below, and can instead be set to the price-space system (29). The choice changes only what each restart minimises. Selection (34) and the acceptance gate (36) always run on the IV-space error J_{iv} , so under the default objective the optimizer minimises the very error that then selects and gates.

IV-space (default in-engine objective; selection and gate). Invert each model price back to a Black vol via (15) and compare to the market vol:

$$\begin{aligned} \sigma_n^{\text{mod}}(\Psi) &= \mathcal{I}(C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n); S_{\text{ref}}, K_n^*, T_n, w_n), \\ \varepsilon_n^{\text{iv}}(\Psi) &= \sigma_n^{\text{mod}}(\Psi) - \hat{\sigma}_n, \end{aligned} \quad (27)$$

with objective

$$J_{\text{iv}}(\Psi) = \left(\frac{1}{|\mathcal{N}'|} \sum_{n \in \mathcal{N}'} \varepsilon_n^{\text{iv}}(\Psi)^2 \right)^{1/2}, \quad \mathcal{N}' = \{n : \sigma_n^{\text{mod}}(\Psi) \text{ inverts finitely}\}. \quad (28)$$

J_{iv} is measured in volatility points; it is the natural surface-fit metric and, unlike J_{px} , is not inflated by cheap deep-OTM wings (a unit price error on a near-worthless contract is a large *relative* price error but a small vol error).

Price-space (alternative in-engine objective). For each target the market value is the Black price at the quoted vol, $C_n^{\text{mkt}} = C_B(\hat{\sigma}_n; S_{\text{ref}}, K_n^*, T_n, w_n)$, and the relative-price residual is

$$\varepsilon_n^{\text{px}}(\Psi) = \frac{C^{\text{mod}}(\Psi; S_{\text{ref}}, K_n^*, T_n, w_n) - C_n^{\text{mkt}}}{C_n^{\text{mkt}}}. \quad (29)$$

The associated objective is the root-mean-square

$$J_{\text{px}}(\Psi) = \left(\frac{1}{|\mathcal{N}|} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{px}}(\Psi)^2 \right)^{1/2}. \quad (30)$$

The per-target residual $\varepsilon_n^{\text{px}}$ is the (relative-price) error each QuantLib helper exposes. Levenberg–Marquardt [7] is a sum-of-squares method, so it drives the squared residual norm $\sum_n (\varepsilon_n^{\text{px}})^2 = |\mathcal{N}| J_{\text{px}}(\Psi)^2$ to a local minimum (equivalently J_{px} , with which it shares a minimiser).

5.2. Multi-start local optimization. A single start into a non-convex problem is fragile [9], so we restart from a small, data-seeded set. Let the surface variance level be

$$\hat{v} = \text{clip} \left(\left(\text{median}_{(k,\tau) \in \mathcal{N}} \hat{\sigma}_{k,\tau} \right)^2, [10^{-3}, 0.25] \right), \quad (31)$$

and define the seed set $\mathcal{S} = \{s_1, \dots, s_6\} \subset \mathcal{B}$ of $(v_0, \kappa, \theta, \eta, \rho)$ tuples

$$\mathcal{S} = \begin{cases} (\hat{v}, 1.0, \hat{v}, 0.50, -0.70), & (\hat{v}, 3.0, \hat{v}, 1.00, -0.50), \\ (\hat{v}, 0.5, \hat{v}, 0.30, -0.90), & (\hat{v}, 5.0, \hat{v}, 0.80, -0.60), \\ (0.8\hat{v}, 2.0, 1.2\hat{v}, 0.60, -0.75), & (\hat{v}, 8.0, \hat{v}, 1.20, -0.40). \end{cases} \quad (32)$$

Under Bates each seed is extended by one fixed jump triple $(\lambda, \nu, \delta) = (0.1, -0.1, 0.1)$, so every restart starts from almost no jumps and grows them only if they tighten the fit; the restart count is unchanged. For each seed $s \in \mathcal{S}$ we solve the box-constrained nonlinear least-squares problem

$$\hat{\Psi}_s = \arg \min_{\Psi \in \mathcal{B}} \sum_{n \in \mathcal{N}} \varepsilon_n^{\text{obj}}(\Psi)^2 \quad (\text{LM, started at } \Psi^{(0)} = s), \quad (33)$$

using Levenberg–Marquardt under the box constraint $\mathcal{B}$ (QuantLib `NonhomogeneousBoundaryConstraint`). The per-target residual $\varepsilon_n^{\text{obj}}$ is the configured in-engine objective: by default the IV-space residual $\varepsilon_n^{\text{iv}}$ (27), optionally the price-space residual $\varepsilon_n^{\text{px}}$ (29). The fit retained for the day is the restart with the smallest *IV-space* error,

$$\Psi^* = \arg \min_{s \in \mathcal{S}} J_{\text{iv}}(\hat{\Psi}_s). \quad (34)$$

Ranking by J_{iv} (not J_{px}) removes the dependence on any single arbitrary guess and selects the economically best surface fit.

6. Acceptance gate

Ψ^* is accepted only if it fits the surface tightly *and* lies in the interior of the box. Let the boundary-pegging predicate, with tolerance fraction b_{tol} , be

$$\text{PEGGED}(\Psi) = \bigvee_{j=1}^{\dim \Phi} \left[(\Psi_j - \ell_j) < b_{\text{tol}} (h_j - \ell_j) \vee (h_j - \Psi_j) < b_{\text{tol}} (h_j - \ell_j) \right]. \quad (35)$$

The disjunction runs over the $\dim \Phi$ diffusive coordinates for both models. Under Bates the three jump coordinates (ν, δ, λ) are exempt from the predicate: $\lambda \rightarrow 0$ is a legitimate collapse to Heston rather than the optimizer hitting a wall, and when $\lambda \approx 0$ the jump shape (ν, δ) is unidentified and may rest on a bound without meaning. The jump triple is still reported. The Feller quantity (4)

likewise stays the diffusive quantity under Bates and never gates. The day is accepted iff

$$\boxed{J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}} \quad \text{and} \quad \neg \text{PEGGED}(\Psi^*)}. \quad (36)$$

The first clause is the IV-space gate ($\varepsilon_{\text{iv}} \approx 2$ vol points). The second rejects boundary fits: a parameter resting on a bound is the optimizer hitting a wall, not an identified value, so it is treated as a non-fit regardless of J_{iv} . The Feller quantity (4) is reported but never gates, since short-tenor SPX fits can violate it while remaining otherwise sound. A rejected or too-thin day yields no parameter row.

Why an IV-space gate. An earlier version gated on J_{px} directly. On a pooled day rich in deep-OTM wings, J_{px} is dominated by those cheap contracts and rejects fits that are excellent in vol points. Replacing the gate metric with J_{iv} aligns the acceptance criterion with the units the surface is quoted in; under it the genuine pooled-surface fit is ≈ 0.8 – 1.4 vol points, and the residual rejections are boundary-pegged κ or ρ (a separate identification lever), not fit-quality failures.

7. Output and diagnostics

Every attempted day produces output. Results are routed by model and objective into one directory, `results/⟨model⟩/calibrations/⟨objective⟩/`, so a Heston run never shares a file with a Bates run, nor an IV-space run with a price-space one. That directory holds three artefacts: a parameter table of accepted days (`calibrations.csv`), a rejection table of dropped days (`rejections.csv`), and a folder of per-day repricing diagnostics (`calibration_tests/`). The two tables partition the attempted days. An accepted day writes one row to the parameter table and nothing to the rejection table; a rejected day does the reverse.

Accepted days: the parameter table. An accepted day appends one row, keyed by the date d , to `calibrations.csv`. The row carries the rate-stamped inputs, the fitted vector, and fit and coverage diagnostics:

$$\begin{aligned} & (d; S_{\text{ref}}, r, g; \theta^*, \kappa^*, \rho^*, \eta^*, v_0^*; \mathcal{F}(\Psi^*); \\ & J_{\text{iv}}(\Psi^*), J_{\text{px}}(\Psi^*), |\mathcal{N}|; |\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|, \sum_i q_i; \\ & \min_i S_i, \max_i S_i, R, \text{HIGH_MOVE}; t_{\text{calc}}). \end{aligned} \quad (37)$$

The blocks are: the reference spot (18) and the day's rates; the fitted parameters Ψ^* with their Feller quantity (4); the two objective values (28) (30) and the helper count $|\mathcal{N}|$; the surface coverage (kept maturities, distinct strikes, populated cells) and the day's total traded volume; the intraday spot range with its flag (19); and t_{calc} , the latest quote timestamp on the surface. A boolean accept flag (always true in this file) is stored alongside. Under Bates the jump triple $(\lambda^*, \nu^*, \delta^*)$ is inserted after v_0^* and nothing else changes. Because the calibration is once-daily, these rows accumulate into the one file, rewritten in date order at the end of each run.

Rejected days: the rejection table. A day dropped at any stage appends one row, also keyed by d , to `rejections.csv`, recording why:

$$(d; \text{REASON}, \text{DETAIL}; J_{\text{iv}}; |\mathcal{T}|, \#\text{strikes}, |\mathcal{N}|). \quad (38)$$

The REASON is one of six categories, in pipeline order: **no_trades** (nothing survives the pre-filter (17)), **no_rate** (no r or g on or before d), **thin** (the coverage requirement (25) fails), **pegged** (the fit clears the IV gate but a diffusive parameter is boundary-pegged (35)), **iv_miss** ($J_{iv}(\Psi^*) > \varepsilon_{iv}$), and **no_fit** (no restart returned a finite fit). **DETAIL** is the human-readable string. The IV error and the coverage counts are filled in where the day reached the stage that computes them, and left blank otherwise: a **thin** day has coverage counts but no J_{iv} , a **no_trades** day has neither. The two tables together cover every attempted day, so the pegged-versus-thin-versus-miss split is auditable directly.

Per-day repricing diagnostics. Each accepted day also writes one file to **calibration.tests/**, holding the day’s repriced surface contracts. One representative trade per populated cell (the highest-volume one, matching the pivot (23)) is valued under Ψ^* . The valuation uses each contract’s *original* (S_i, K_i), not (S_{ref}, K^*): the structural parameters are spot-independent, and the honest comparison is against the actual trade. Each row records the original spot and strike, the snapped K^* , the fitted parameters, the market implied volatility σ_i , the Black price (9) at that vol, and the model price (the Heston price (10), or a Bates price column (14) under Bates).

8. Constants

Symbol	Code name	Value	Role
$\tau_{\min}$	MIN_DTM	14	minimum days to maturity
$\tau_{\max}$	MAX_DTM	730	maximum days to maturity
$\underline{m}$	OTM.MONEYNESS.FLOOR	0.6	minimum OTM ratio moneyness
$\bar{m}$	OTM.MONEYNESS.CUTOFF	0.98	maximum OTM ratio moneyness
$R_{\max}$	MAX.MOVE.PCT	0.03	intraday-range flag threshold
Δ	STRIKE_GRID	5.0	SPX strike lattice for K^* snap
$\bar{N}_t$	MAX_NT	20	maturities kept (by volume)
$\underline{n}_k$	MIN_NK	2	minimum distinct strikes per wing
$\bar{n}_k$	MAX_NK	40	strikes kept per wing
M_t	MIN_MATS	3	minimum maturities in surface
M_k	MIN_STRIKES	5	minimum strikes in surface
M_{cell}	MIN_CELLS	12	minimum populated cells $ \mathcal{N} $
ε_{iv}	IV_RMSE_ACCEPT	0.02	IV-space acceptance gate (vol pts)
b_{tol}	BOUND_TOL	10^{-3}	boundary-pegging tolerance fraction

TABLE 1. Surface, selection, and gate constants.

	θ	κ	η	ρ	v_0		λ	ν	δ
lower ℓ	10^{-4}	10^{-2}	10^{-2}	-0.999	10^{-4}	lower ℓ	0.0	-0.5	10^{-3}
upper h	1.0	20.0	2.0	0.5	1.0	upper h	5.0	0.2	0.5

TABLE 2. Box bounds $\mathcal{B}$ (26), in the **model.params()** order. Left: the five Heston coordinates ($\theta, \kappa, \eta, \rho, v_0$); the ρ upper bound is kept slightly positive (equity leverage implies $\rho < 0$) and η is capped at 2.0 (SPX vol-of-vol ~ 0.3 –1.2). Right: the Bates jump bounds (λ, ν, δ) appended when $\Psi = \Theta$; the λ floor is exactly 0 (the fit may collapse to pure Heston) and ν allows more negative room since equity jumps skew down.

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Appendix A. Algorithm

Algorithm 1 states the per-day routine in the active vector Ψ ; the driver applies it to every trading day and accumulates accepted rows into a single file (§7). Heston is the default ($\Psi = \Phi$). The Bates fit is the same algorithm with $\Psi = \Theta$: the engine is the **BatesEngine** and the box $\mathcal{B}$ is the dim Θ -dimensional one, while the pegging test (35) still ranges over the dim Φ diffusive coordinates only.

Algorithm 1 Per-day calibration

Require: day d of OTM trades; rates $r(d), g(d)$; constants of Table 1

- 1: $\mathcal{D} \leftarrow \{i : \sigma_i > 0, \tau_i \geq \tau_{\min}\}$ (pre-filter (17))
- 2: **if** $r(d)$ or $g(d)$ undefined on/before d **then**
- 3: **return** $\emptyset$ (no rate)
- 4: **end if**
- 5: $S_{\text{ref}} \leftarrow (\sum q_i S_i) / (\sum q_i)$;
 $R \leftarrow \max_i S_i / \min_i S_i - 1$;
 $\text{HIGH_MOVE} \leftarrow \mathbf{1}\{R > R_{\max}\}$
- 6: $K_i^* \leftarrow \Delta \text{round}(m_i S_{\text{ref}} / \Delta)$ with $m_i = K_i / S_i$ (re-strike (20))
- 7: $\mathcal{T} \leftarrow \text{top-}\bar{N}_t$ admissible maturities by volume ($\geq \underline{n}_k$ strikes per wing);
 build $\mathcal{K}_\tau$ (maximum $\bar{n}_k$ per wing);
 pivot to $\{\hat{\sigma}_{k,\tau}\}$, cells $\mathcal{N}$
- 8: **if** coverage (25) fails **then**
- 9: clear stale diagnostics; **return** $\emptyset$
- 10: **end if**
- 11: $\hat{v} \leftarrow \text{clip}(\text{median}(\hat{\sigma})^2, [10^{-3}, 0.25])$; build seeds $\mathcal{S}$ (31) (32)
- 12: **for** $s \in \mathcal{S}$ **do**
- 13: $\hat{\Psi}_s \leftarrow \text{LM minimiser of } \sum_n (\varepsilon_n^{\text{obj}})^2 \text{ on } \mathcal{B} \text{ from } s$ (33)
- 14: compute $J_{\text{iv}}(\hat{\Psi}_s)$ (28)
- 15: **end for**
- 16: $\Psi^* \leftarrow \arg \min_s J_{\text{iv}}(\hat{\Psi}_s)$ (select (34))
- 17: **if** $J_{\text{iv}}(\Psi^*) \leq \varepsilon_{\text{iv}}$ **and** $\neg \text{PEGGED}(\Psi^*)$ **then**
- 18: write repriced diagnostics; **return** $\text{row}(d, S_{\text{ref}}, r, g, \Psi^*, \mathcal{F}, J_{\text{iv}}, J_{\text{px}}, \dots)$
- 19: **else**
- 20: clear stale diagnostics; **return** $\emptyset$ (rejected)
- 21: **end if**
